

```

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

df <- read_csv("data/data.csv")

## Rows: 15 Columns: 1
## -- Column specification -----
## Delimiter: ","
## dbl (1): agewed
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

#data for computing the differenet confidence intervals
n      <- nrow(df)                # sample size
mean_age <- mean(df$agewed, na.rm=TRUE) # sample mean
sd_age   <- sd(df$agewed, na.rm=TRUE)  # sample SD
se_age   <- sd_age / sqrt(n)          # standard error
dfree    <- n - 1

#verify these work
n; mean_age; sd_age; se_age; dfree

## [1] 15
## [1] 24.06667
## [1] 8.075241
## [1] 2.085018
## [1] 14

# finding t values for each confidence interval we want to do (95, 80 and 50)
t95 <- qt(0.975, dfree) # 95%
t50 <- qt(0.75, dfree)  # 50%
t80 <- qt(0.90, dfree)  # 80%

t95; t50; t80

## [1] 2.144787
## [1] 0.6924171
## [1] 1.34503

ci95 <- mean_age + c(-1,1) * t95 * se_age
ci50 <- mean_age + c(-1,1) * t50 * se_age
ci80 <- mean_age + c(-1,1) * t80 * se_age

```

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# display each interval  
ci95; ci50; ci80
```

```
## [1] 19.59475 28.53859
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## [1] 22.62296 25.51037
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## [1] 21.26225 26.87108
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Interpretation for 95%: We are 95% confident that the true average age at first marriage in 1972 for all US adults is between 19.59475 and 28.53859 years old.

Interpretation for 80%: We are 80% confident that the true average age at first marriage in 1972 for all US adults is between 22.62296 and 25.51037 years old.

Interpretation for 50%: We are 50% confident that the true average age at first marriage in 1972 for all US adults is between 21.26225 and 26.87108 years old. ““